EARTHQUAKE INSPECTION - RESULTS

Machine learning + quantum optimization + parallel verification

1 Formulation

Binary x_i indicates inspection. Maximize sum(yhat_i*x_i), subject to sum(x_i)=3 and sum(t_i*x_i)<=10. There are C(8,3)=56 triples; 35 meet the time limit.

2 Working solution across all three domains

OLS with intercept matches the linear synthetic generator (seed 63, N=120). Independent 80/20 split (seed 7): MAE 0.4137, RMSE 0.5260, R2 0.9632. Final fit uses all 120 records: yhat=5.129735+0.297040*d+0.080070*a+3.976314*l-0.125008*r.

Location	d (1000/km2)	a (years)	l (0-1)	r (km)	Time (h)	Predicted damage
L1	14	35	0.8	6	4	14.5218
L2	9	50	0.3	12	2	11.4994
L3	18	20	0.6	25	3	11.3384
L4	6	55	0.9	8	5	13.8944
L5	16	45	0.5	10	4	14.2236
L6	11	30	0.7	18	3	11.3326
L7	4	40	0.2	30	3	6.5657
L8	12	25	0.4	15	3	10.4114

Let s=b0+2b1+4b2+8b3. Minimize E=-sum(yhat_i*x_i)+A(sum(x_i)-3)^2+B(sum(t_i*x_i)+s-10)^2. A=B=1+sum(abs(yhat_i))=94.7873; a unit violation costs more than the reward range. Nonnegative slack prevents over-time solutions. The QUBO uses 12 qubits; all 4096 energies were checked against its expanded polynomial and Ising form. QAOA: |+>^12, alternating exp(-i*gamma*E/A) and Rx(2*beta); depths 1,2,3,4,6, three starts each, 11465 optimizer evaluations total; chosen p=6. Exact expectations train angles. The final elementary H/Rz/CX/Rx circuit is simulated independently (fidelity > 1-1e-10) and sampled for 4096 shots, seed 2026. Circuit: circuit.qasm; coefficients/settings: qubo.json.

A spawn multiprocessing Pool with 2 workers evaluates each triple once; a sequential pass confirms the optimum. Runtimes below are single-run wall times: greedy sorting/evaluation; quantum encoding, validation, optimization and sampling; parallel pool startup/evaluation/reduction. Common data/ML time is 0.005073 s and is excluded. Process overhead dominates this tiny workload.

3 Measured comparison

Method	Selected locations	Damage	Hours	Time <=10?	Seconds
Greedy top-3	L1,L4,L5	42.6398	13	No	0.000007
QAOA best feasible	L1,L2,L5	40.2448	10	Yes	9.812743
QAOA modal state	L1,L2,L8	36.4325	9	Yes	9.812743
Parallel exhaustive	L1,L2,L5	40.2448	10	Yes	0.075279

4 Interpretation and limitations

The best feasible QAOA sample matches exhaustive search. Damage gap: 0.000000 (0.0000%). Optimum selection: 29/4096 shots (0.71%); feasible shots: 36.91%. The modal full bitstring selects L1,L2,L8; the modal selection after combining slack outcomes is L7,L8 (feasible: False). Observing the optimum among 4096 shots does not establish concentration or quantum advantage.

1. Synthetic linear data and a small held-out set do not establish real earthquake accuracy; prediction uncertainty may change the selected triple. 2. Shallow QAOA and a bounded local optimizer offer no convergence guarantee; classical statevector memory grows exponentially. No hardware noise is modeled.

References

1. NumPy: least-squares regression: https://numpy.org/doc/stable/reference/generated/numpy.linalg.lstsq.html
2. Farhi, Goldstone & Gutmann (2014), QAOA: https://arxiv.org/abs/1411.4028
3. Python: multiprocessing process pools: https://docs.python.org/3/library/multiprocessing.html#using-a-pool-of-workers